TIPPSS SafeGuard-PQ
Consolidated Technical Report

Agentic Post-Quantum Cybersecurity Assessment & Migration Planning
for Clinical Edge IoT and AI Healthcare Systems

Author:	Pavan Gemmanjunath
Contact:	pavan5119m@gmail.com
Repository:	github.com/pavangemmanjunath-cell/QuantumShift_TIPPSS
Standards:	NIST FIPS 203 / 204 / 205 · IEEE 2933-2024 TIPPSS
Frameworks:	HIPAA · OWASP LLM Top-10 · Kirkpatrick SA 1983
Novel Element:	QIRO — Quantum-Inspired Remediation Optimizer

1. Problem Statement

Healthcare organisations operate clinical Internet of Things (IoT) networks comprising infusion pumps, ECG monitors, patient wearables, edge gateways, and AI-assisted EHR systems. All of these systems transmit and store sensitive patient data using classical public-key cryptography — RSA-2048, ECDSA-P256, ECDH — that will be rendered mathematically insecure by Shor's algorithm running on a cryptographically relevant quantum computer, with expert consensus placing the threat window at 2030–2035.

1.1 The Harvest-Now, Decrypt-Later (HNDL) Threat

Adversaries are already capturing encrypted clinical network traffic today with the explicit intent of decrypting it when quantum hardware matures. Since healthcare data must remain confidential for 15–50 years under HIPAA and GDPR, records currently protected by RSA or ECDSA are already exposed — the harvest window has begun.

1.2 Six Identified Security Gaps

- Quantum-vulnerable cryptography: RSA, ECDSA, ECDH, deprecated TLS across clinical infrastructure — no automated PQ readiness inventory exists.
- PHI leakage in AI systems: Patient names, MRNs, diagnoses, and email addresses appearing in clinical AI prompt logs — invisible to classical scanners.
- Prompt injection against clinical AI: OWASP LLM Top-10 attack patterns (instruction override, data exfiltration) targeting clinical AI assistants.
- Exposed credentials in device telemetry: API keys, AWS access keys, and bearer tokens embedded in medical device debug logs.
- Weak medical IoT device identity: Missing certificates, failed authentication, repeated login failures undermining Trust and Identity pillars.
- No post-quantum migration path: No existing tool maps findings to NIST FIPS 203/204/205 and produces a budget-constrained, downtime-aware roadmap.

1.3 Quantified Risk Exposure

Metric	Value	Source
Average healthcare breach cost	$9.77 million	IBM Cost of a Data Breach 2024
Healthcare breaches (2023)	725+ incidents / 133M+ records	HHS Office for Civil Rights
Quantum threat timeline	2030–2035	NIST IR 8547 / NSA CNSA 2.0
Federal PQC migration deadline	2035 (RSA/ECC deprecated after 2030)	NIST IR 8547
Healthcare data retention	15–50 years	HIPAA / national regulations

2. Solution Overview

TIPPSS SafeGuard-PQ is a sequential agentic cybersecurity assessment and migration planning platform. It evaluates clinical IoT systems and AI healthcare applications against NIST FIPS 203/204/205 post-quantum standards and the IEEE 2933-2024 TIPPSS framework through a nine-agent sequential pipeline, then produces a constraint-aware remediation roadmap using the novel Quantum-Inspired Remediation Optimizer (QIRO).

2.1 Nine-Agent Assessment Pipeline

Agent	Function	TIPPSS Pillar	Standards
1 — Input Validator	Schema validation · field normalisation	All	—
2 — Crypto Classifier	Classify TLS: PQ-safe / classical / deprecated	Protection, Security	FIPS 203/204/205
3 — PQ Benchmark	NIST compliance · HNDL rule (lifetime ≥ 10 yr)	Protection, Privacy	FIPS 203/204/205
4 — IoT Device Identity	Certificate · auth status · failed logins	Trust, Identity, Safety	IEEE 2933-2024 §6.2
5 — AI Prompt Privacy	PHI detection · prompt injection scanning	Privacy, Security	IEEE 2933-2024 §6.3 · HIPAA
6 — Secret Exposure	API keys · AWS · GitHub tokens in device logs	Protection, Security	OWASP
7 — Safety Risk	Autonomous dosing · alarm overrides	Safety, Trust	IEEE 2933-2024 §6.5
8 — TIPPSS Scorer	**Six-pillar scoring · composite risk verdict**	**All six pillars**	**IEEE 2933-2024 §6.1–6.6**
9 — QIRO Optimizer	QUBO binary optimisation · migration roadmap	All pillars	Kirkpatrick 1983

2.2 Novel Contribution — QIRO

QIRO (Quantum-Inspired Remediation Optimizer) converts every detected finding into a candidate remediation action and frames the selection problem as a Quadratic Unconstrained Binary Optimisation (QUBO) — the same problem class solved by quantum annealers. It then solves this on a classical CPU using Simulated Annealing (Kirkpatrick, Gelatt, Vecchi, 1983), selecting the maximum-benefit feasible portfolio under the hospital's budget and downtime constraints. No existing post-quantum migration tool applies this formulation.

3. System Architecture Diagram

The platform implements a three-tier architecture. Tier 1 (Streamlit, port 8501) handles scenario selection, file upload, feature extraction, and result visualisation. Tier 2 (Sequential Agentic Framework, port 5678) orchestrates the nine-agent pipeline. Tier 3 persists timestamped results as JSON and CSV.

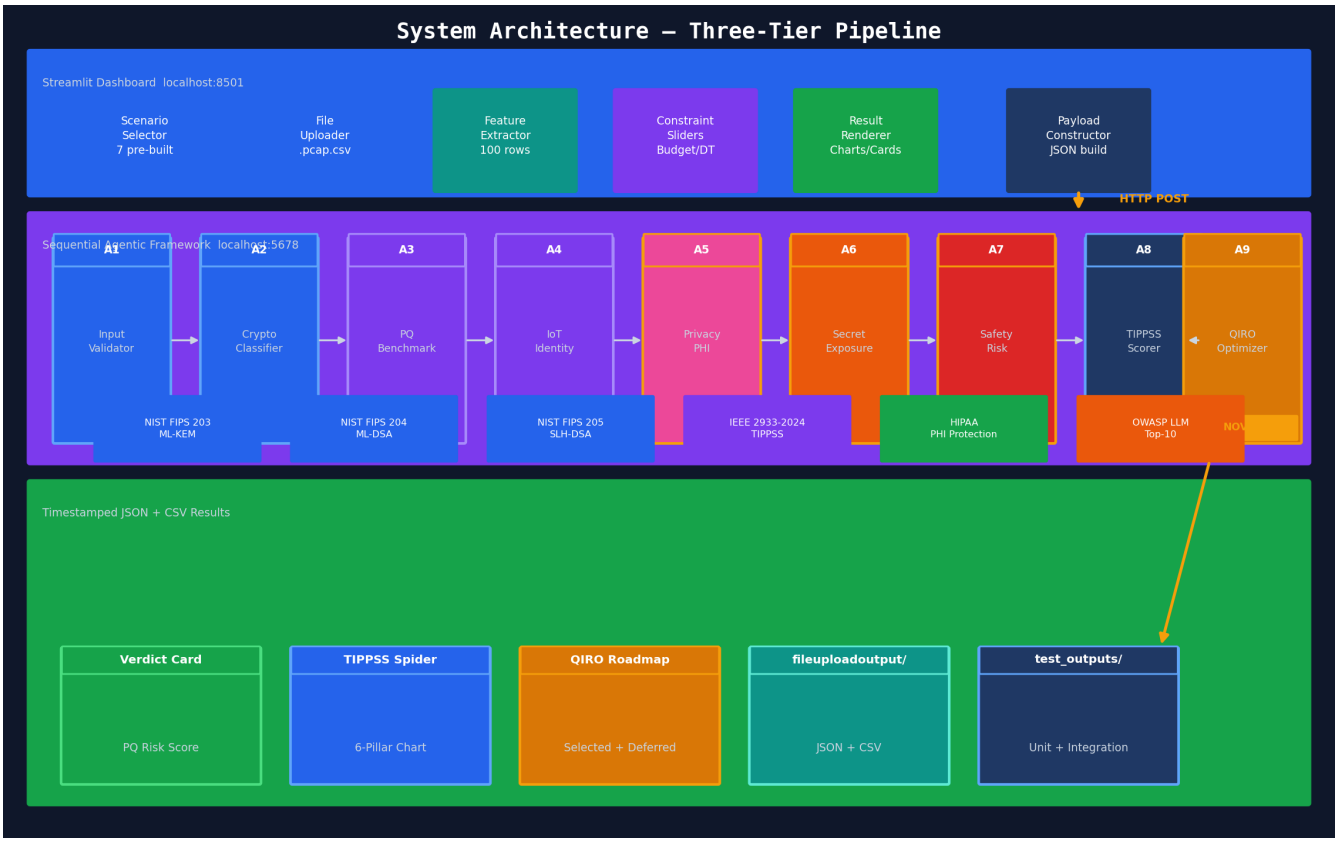


Figure 1: Three-tier system architecture — Presentation, Orchestration, and Output layers.

4. End-to-End Data Flow

Data enters as a JSON payload (from pre-built scenarios or uploaded files), passes sequentially through nine agents, and exits as a verdict, six-pillar TIPPSS score, risk gauge, and QIRO migration roadmap. The gold HTTP POST arrow represents the single webhook call from the dashboard to the agentic framework.

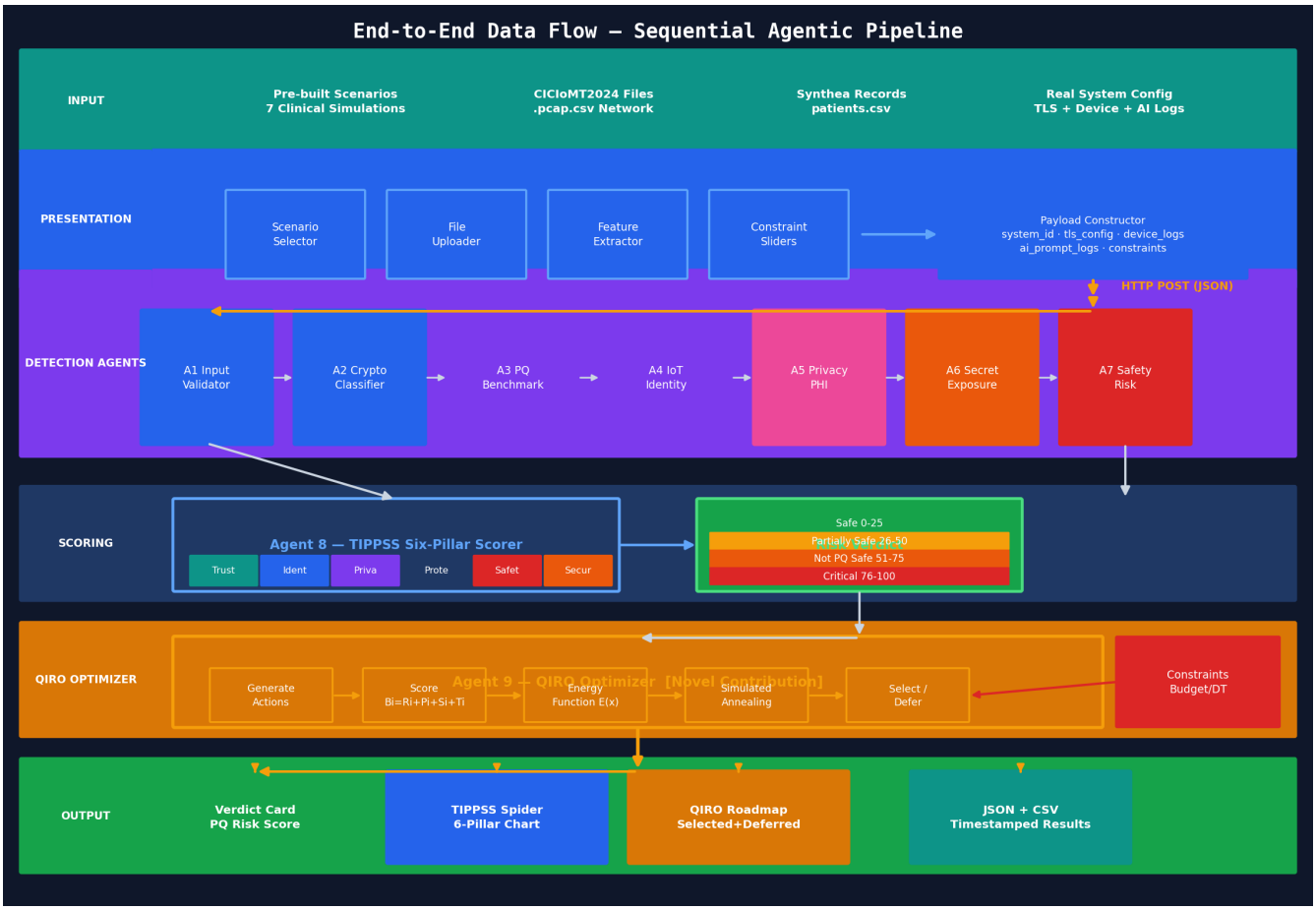


Figure 2: Swimlane data flow from input sources through all agents to output components.

5. Network Topology & Threat Model

The threat model maps six attack classes — HNDL adversary, quantum computation, prompt injection, ARP spoofing / MQTT malformation, secret exposure, and DoS/DDoS — against the hospital clinical network perimeter. TIPPSS SafeGuard-PQ is positioned as the assessment layer covering all six TIPPSS pillars across the clinical IoT estate.

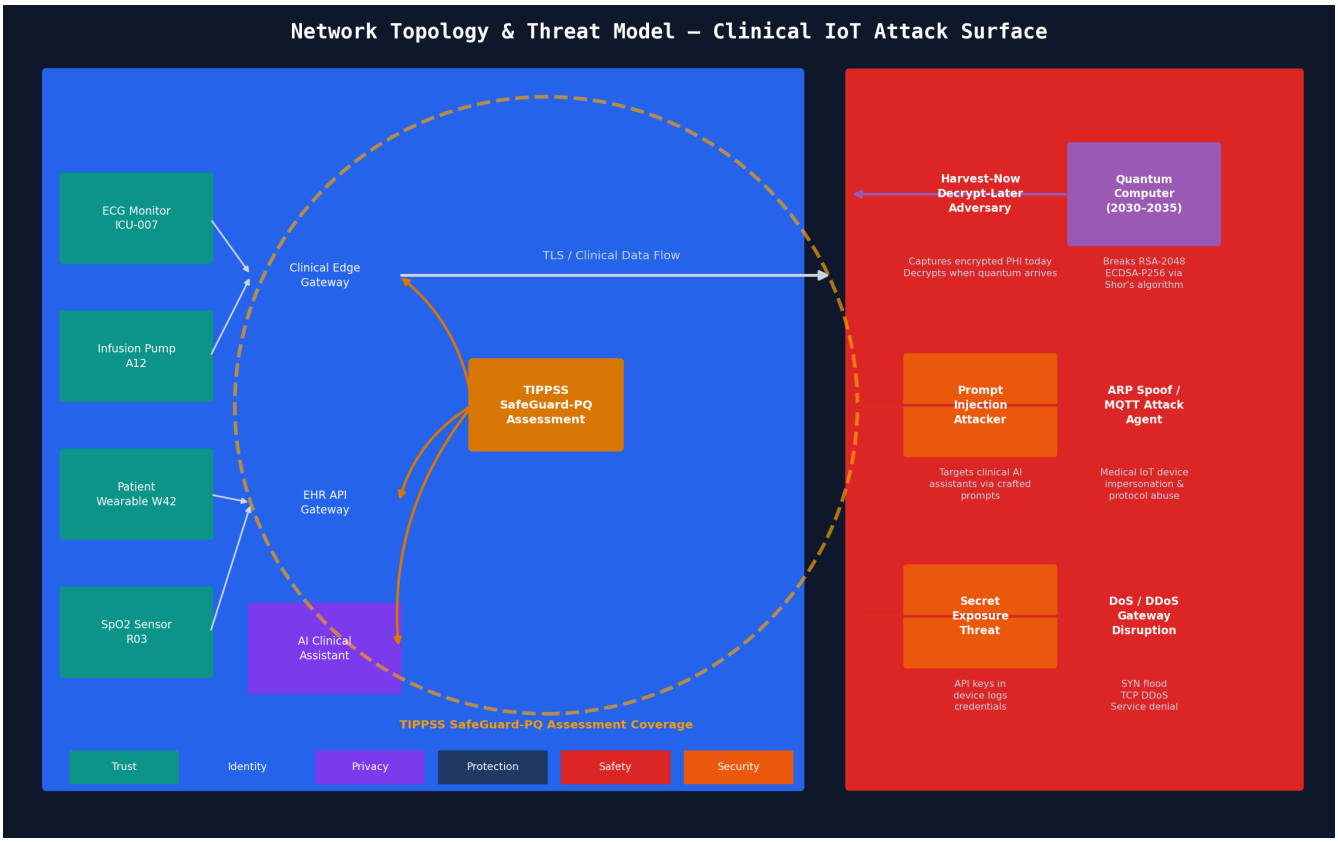


Figure 3: Clinical network topology with threat actors, attack vectors, and TIPPSS assessment coverage.

6. QIRO Optimisation Diagram

Left panel: full mathematical formulation of QIRO — decision variable, benefit score Bi = Ri + Pi + Si + Ti, objective function, penalty-based energy function E(x), Simulated Annealing algorithm, and QUBO/Ising connection. Right panel: ICU gateway use case showing seven candidate actions with benefit scores, budget tracking, phased migration roadmap, and selected/deferred action split.



Figure 4: QIRO mathematical formulation (left) and ICU gateway use case walkthrough (right).

7. Test Results & Empirical Validation

All 8/8 file upload tests passed against the live sequential agentic pipeline. Default constraints: budget $100,000, downtime 40 hours. The benign baseline correctly returns a Safe verdict (score 0, zero findings), confirming correct discrimination from attack traffic.

File	Attack Type	Verdict	Score	Findings	QIRO Selected	QIRO Deferred
ARP_Spoofing_test.pcap.csv	ARP Spoofing	Not PQ Safe	61/100	7	4	3
Benign_test.pcap.csv	Baseline	Safe	0/100	0	0	0
MQTT-Malformed_Data.pcap.csv	MQTT Malformed	Not PQ Safe	61/100	7	4	3
Recon-Port_Scan.pcap.csv	Port Scan	Not PQ Safe	61/100	7	4	3
TCP_IP-DoS-SYN.pcap.csv	SYN DoS	Not PQ Safe	61/100	7	4	3
TCP_IP-DDoS-TCP.pcap.csv	TCP DDoS	Not PQ Safe	61/100	7	4	3
patients.csv (Synthea)	PHI Simulation	Safe (TLS)	0/100	8 PHI	8	0
conditions.csv (Synthea)	Clinical Conds.	Safe	0/100	0	0	0

8. Standards Compliance Matrix

Component	Standard	Clause / Reference	Status
Agent 3 PQ Benchmark	NIST FIPS 203	ML-KEM key encapsulation	Implemented
Agent 3 PQ Benchmark	NIST FIPS 204	ML-DSA digital signatures	Implemented
Agent 3 PQ Benchmark	NIST FIPS 205	SLH-DSA hash-based signatures	Implemented
Agent 4 IoT Identity	IEEE 2933-2024	Clause 6.2 — Trust and Identity	Implemented
Agent 5 AI Privacy	IEEE 2933-2024	Clause 6.3 — Privacy	Implemented
Agent 5 PHI Detection	HIPAA 45 CFR §164.514	Minimum necessary — AI inputs	Implemented
Agent 7 Safety Risk	IEEE 2933-2024	Clause 6.5 — Safety	Implemented
Agent 8 TIPPSS Scorer	IEEE 2933-2024	Clauses 6.1–6.6 — All six pillars	Implemented
QIRO Algorithm	Kirkpatrick et al. 1983	Simulated Annealing (Science 220)	Implemented
QIRO Formulation	**QUBO / Ising model**	**Binary quadratic optimisation**	**Novel**

This document was prepared for technical evaluation of the TIPPSS SafeGuard-PQ system. All datasets are publicly available research data (CICIoMT2024 — University of New Brunswick; Synthea — MITRE Corporation, Apache 2.0). No real patient data was used.